CSE STUDY HUB

C++ Programming

UNIT 2 · STL and C++20 Ranges

What this unit is about

This unit builds understanding of **STL and C++20 Ranges**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Standard Template Library (STL): containers, iterators, algorithms and function objects

2. Sequence containers: vector, deque, list, array and forward_list

3. Associative containers: set, multiset, map and multimap

4. Unordered containers: unordered_set and unordered_map; hashing

5. Iterators: input, output, forward, bidirectional and random-access

6. Common algorithms: sort, find, count, transform, accumulate and lower_bound

7. C++20 ranges, views, pipelines and lazy evaluation

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 2 NOTES · PART 1

Standard Template Library (STL): containers, iterators, algorithms and function objects

Present Standard Template Library (STL): containers, iterators, algorithms and function objects as an ordered workflow. State the preconditions, list the major steps, explain what each step produces, and mention how errors are detected or handled. Use a small worked example or flowchart in the exam.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Sequence containers: vector, deque, list, array and forward_list

Sequence containers: vector, deque, list, array and forward_list is a syllabus concept in C++ Programming. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Associative containers: set, multiset, map and multimap

Associative containers: set, multiset, map and multimap is a syllabus concept in C++ Programming. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on STL and C++20 Ranges, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 2 NOTES · PART 2

Unordered containers: unordered_set and unordered_map; hashing

A hash function maps a key to a table index. Collisions are handled by chaining or open addressing. Performance depends on a good hash function, load factor and resizing policy.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Iterators: input, output, forward, bidirectional and random-access

Draw a compass and track each turn. Distinguish total distance travelled from net displacement. For shadow problems, the shadow points opposite the sun direction.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on STL and C++20 Ranges, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 2 NOTES · PART 3

Common algorithms: sort, find, count, transform, accumulate and lower_bound

Sorting arranges records by a key. Compare algorithms using time, auxiliary space, stability and in-place behaviour. Merge sort is O(n log n); quicksort averages O(n log n) but can degrade to $O(n^2)$.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

C++20 ranges, views, pipelines and lazy evaluation

C++20 ranges, views, pipelines and lazy evaluation is a syllabus concept in C++ Programming. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on STL and C++20 Ranges, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Standard Template Library (STL): containers, iterators, algorithms and function objects** — know the definition, core working and one example.

• **Sequence containers: vector, deque, list, array and forward_list** — know the definition, core working and one example.

• **Associative containers: set, multiset, map and multimap** — know the definition, core working and one example.

• **Unordered containers: unordered_set and unordered_map; hashing** — know the definition, core working and one example.

• **Iterators: input, output, forward, bidirectional and random-access** — know the definition, core working and one example.

• **Common algorithms: sort, find, count, transform, accumulate and lower_bound** — know the definition, core working and one example.

• **C++20 ranges, views, pipelines and lazy evaluation** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from STL and C++20 Ranges. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Standard Template Library (STL): containers, iterators, algorithms and function objects and explain its importance.

Q2. Explain the working/steps involved in Sequence containers: vector, deque, list, array and forward_list.

Q3. Differentiate between two related concepts from STL and C++20 Ranges.

Q4. Give one practical application of C++20 ranges, views, pipelines and lazy evaluation and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of STL and C++20 Ranges.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.